

SARYKOL, Kazakhstan

WMO: 288670

Lat: **53.327N**Lon: **65.524E**Elev: **216**StdP: **98.76**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-28.3	-34.9	0.1	-31.4	-31.7	0.2	-28.2	15.1	-10.3	12.4	-10.8	2.0	20	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.5	19.5	29.6	19.0	27.9	18.3	20.8	29.3	19.9	28.0	19.0	26.7	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.9	26.2	16.5	12.1	25.0	15.5	11.3	24.0	61.2	29.2	57.7	28.0	54.7	26.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.4	9.0	-33.7	34.6	3.7	1.9	-36.3	35.9	-38.5	37.0	-40.6	38.1	-43.2	39.4		
(5)			-33.9	22.5	3.4	1.5	-36.3	23.6	-38.3	24.5	-40.2	25.3	-42.7	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-16.4	-15.3	-6.9	5.3	13.8	18.8	19.8	18.4	12.4	4.6	-5.6	-13.1
	DBStd	14.41	6.76	6.47	5.53	6.03	4.96	4.24	3.60	4.27	4.82	4.95	7.02	6.91
	HDD10.0	3628	819	709	524	163	21	1	0	1	27	178	469	717
	HDD18.3	5781	1078	942	783	391	156	45	27	53	188	424	719	975
	CDD10.0	1100	0	0	0	22	137	266	303	262	99	12	0	0
	CDD18.3	210	0	0	0	0	14	59	71	56	9	0	0	0
	CDH23.3	2102	0	0	0	7	185	622	655	525	107	1	0	0
	CDH26.7	648	0	0	0	1	41	206	196	172	33	0	0	0
Wind	WSAvg	3.4	3.4	3.7	3.6	3.4	3.8	3.4	3.0	2.8	3.0	3.2	3.3	3.6
Precipitation	PrecAvg	352	19	14	18	23	36	42	55	42	25	30	28	21
	PrecMax	499	50	54	53	65	103	99	144	103	56	64	58	47
	PrecMin	182	2	1	1	1	3	4	17	6	3	3	5	4
	PrecStd	78	12	11	13	16	24	27	32	30	15	18	14	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	0.7	7.7	24.0	30.2	33.9	33.6	33.2	30.8	21.3	8.9	1.1
		MCWB	-0.8	-0.8	4.9	14.2	18.0	20.0	20.7	20.2	18.2	13.0	6.0	-0.6
	2%	DB	-2.4	-1.3	3.3	20.5	27.6	31.2	30.7	30.7	26.2	17.9	6.1	-0.5
		MCWB	-3.4	-2.4	1.3	13.1	17.4	18.7	20.3	19.6	16.3	11.1	4.1	-1.6
	5%	DB	-4.0	-3.7	1.7	17.6	25.3	29.1	28.8	28.5	23.1	15.1	4.3	-2.0
		MCWB	-5.0	-4.6	0.0	11.6	16.3	18.5	19.2	18.9	15.0	9.4	2.6	-3.0
	10%	DB	-6.0	-5.9	0.6	14.8	22.9	26.9	27.0	26.3	20.4	12.3	2.3	-3.8
		MCWB	-6.7	-6.8	-0.9	9.9	15.0	17.5	18.5	18.0	13.9	8.0	0.8	-4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.5	4.9	15.7	20.1	21.5	22.2	21.6	19.1	13.5	6.5	-0.3
		MCDB	-0.1	0.5	7.5	22.3	28.3	30.9	30.5	30.6	29.2	20.4	8.4	0.8
	2%	WB	-3.3	-2.3	1.5	13.4	18.2	20.0	21.0	20.4	17.0	11.4	4.2	-1.6
		MCDB	-2.6	-1.2	3.0	20.1	26.0	29.1	29.2	29.1	24.4	17.1	6.1	-0.5
	5%	WB	-4.9	-4.5	0.2	11.6	16.7	19.0	19.9	19.3	15.3	9.7	2.6	-3.0
		MCDB	-4.1	-3.6	1.6	17.3	24.5	27.7	27.4	27.3	22.3	14.6	4.2	-2.0
	10%	WB	-6.7	-6.8	-0.9	10.0	15.2	17.9	19.0	18.2	14.0	8.1	0.9	-4.7
		MCDB	-6.0	-6.0	0.6	14.5	22.3	25.9	26.2	25.5	20.2	12.0	2.2	-3.9
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	7.8	9.5	11.7	11.4	10.4	10.8	10.7	8.6	6.2	7.0
		MCDBR	7.0	7.8	7.1	14.4	14.8	14.2	13.2	14.1	14.6	13.5	7.1	7.0
		MCWBR	6.8	7.3	6.2	9.4	8.4	7.1	6.5	7.0	7.7	8.4	6.0	6.6
	5% WB	MCDBR	7.0	7.6	7.0	13.5	13.9	13.2	12.0	13.0	13.6	12.6	6.9	7.1
		MCWBR	6.8	7.2	6.2	9.3	8.8	7.7	6.8	7.3	8.0	8.3	6.0	6.7
Clear-Sky Solar Irradiance	taub	0.288	0.321	0.372	0.408	0.407	0.403	0.418	0.427	0.374	0.352	0.320	0.288	
	taud	2.065	2.016	1.971	2.132	2.222	2.308	2.263	2.209	2.308	2.264	2.157	2.050	
	Ebn at Noon	649	746	785	817	841	848	826	791	792	717	607	560	
	Edh at Noon	73	107	140	136	131	122	125	125	100	84	68	62	
All-Sky Solar Radiation	RadAvg	1.14	2.27	3.77	4.70	5.77	6.21	5.80	4.80	3.48	1.85	0.90	0.75	
	RadStd	0.12	0.11	0.35	0.49	0.49	0.46	0.45	0.47	0.26	0.26	0.10	0.12	

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page